

such as the remote volume size, which is set to 50MB by default and is suitable unless you're dealing with very large files. For backup retention, selecting 'Smart Backup Retention' is a good starting point, as it automatically manages backup versions based on age and available space. Email notifications can also be set up later once the job is created. After reviewing the settings, click on *Save* to complete the configuration. The new job will now appear on the main dashboard, ready to run. You can click on *Run now* to perform the first backup immediately. With this, your backup job is fully configured and ready to protect your data.

Running and testing backup

Once the backup job has been created, it is important to run it manually at least once to verify that everything works as expected.

Start the first backup: From the Duplicati dashboard, locate your newly created backup job. Click the dropdown arrow next to it and select 'Run now' to start the backup immediately.

Duplicati will begin scanning the selected folders and start uploading the encrypted backup to the Ubuntu server over SFTP. The process may take a few

minutes depending on the size of your files and the speed of the network.

Monitor the backup process: As the backup runs, you can click on the job to monitor its progress in real time. Once the process is complete, Duplicati will display a confirmation message indicating that the backup finished successfully. You can also view useful details such as the total amount of data backed up, the number of files processed, the duration of the backup, and any warnings or errors that may have occurred during the operation.

Verify files on the server: To confirm that the backup was received correctly, log in to your Ubuntu VM and navigate to the backup directory:

```
$ cd /home/backupuser/duplicati-backups
$ ls -lh
```

You should see a set of *.zip.aes* files and metadata files. These are the encrypted backup volumes created by Duplicati.

This confirms that your data was securely transmitted and stored.

Restoring files

A backup system is only useful if it can

restore data when needed. To test this, open the Duplicati web interface, click on your backup job name, and select 'Restore files'. When prompted, choose the option to restore from the backup files. This begins the recovery process using your most recent backup.

To select the backup location, set the storage type to SFTP (SSH), then enter the IP address of the Ubuntu machine, such as 192.168.1.6. Use the default port 22, specify the path on the server as */home/backupuser/duplicati-backups*, and provide the username *backupuser* along with the password set for that account on the Ubuntu system. Then enter the passphrase you created and select the files you need to restore.

Duplicati allows you to restore the file either to its original location or to a different folder, which is recommended for first-time testing; select your preferred option and click on *Restore*. Once the restore is complete, navigate to the selected restore location and open the restored file to confirm that it is intact and usable.

This confirms that your backup job is not only storing data correctly but can also recover files when needed. With both backup and restore operations tested successfully, your system is now functionally complete.

Email notifications

It is important to be notified when a backup succeeds, fails, or is completed with warnings. Duplicati allows you to set up automated email notifications for every operation, including backups and restores. When configured globally, this setup applies to all jobs on the machine, so there is no need to repeat it for each backup task. Let's learn how to configure email alerts using a Gmail account.

Open global settings: Open the Duplicati web interface, click on *Settings* in the left sidebar, scroll down to the *Default options* section, and click on *Edit as text*.

Paste email notification

configuration: Copy and paste the

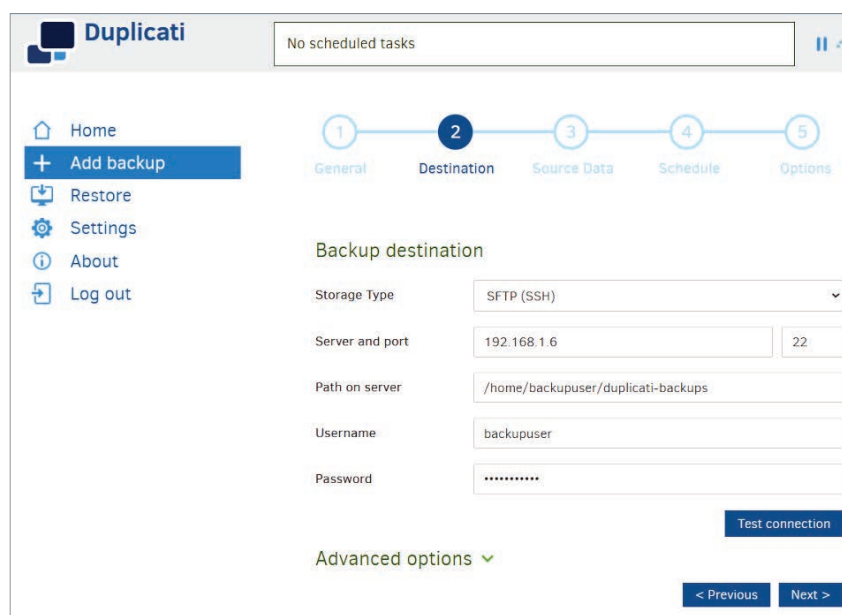


Figure 2: Automation of backups by scheduling